

FAST — National University of Computer and Emerging Sciences*Karachi Campus**Department of Software Engineering***PF LAB TASKS**

Course Code:	C1002
Subject:	Programming Fundamentals
Section:	D & C
Instructor:	Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

Name:	Muhammad Huzaifa Hafeez
Roll No / ID:	26K-3117
Class / Section: D	D
Date of Submission:	28/09/2026

Submission Guidelines

1. Write and compile each program separately in a C compiler (e.g. Code::Blocks, Dev-C++, or an online IDE).
2. Test each program with more than one input where relevant, to confirm all conditions work correctly.
3. Submit all tasks before the given deadline.

TASK # 01 — Temperature Status (*if-else*)

A weather monitoring system takes the temperature in Celsius from the user. Write a C program using if-else to display "Cold" if the temperature is below 15, "Normal" if the temperature is between 15 and 30, and "Hot" if the temperature is above 30.

```
#include <stdio.h>

int main()
{
    int Temp;
    printf("enter temperature");
    scanf("%d",&Temp);
    if(Temp<15)
    {
        printf("Cold");
    }
}
```

```
}  
else if(Temp>=15 && Temp<30)  
{  
    printf("Normal");  
}  
else if(Temp>=30)  
{  
    printf("Hot");  
}  
else  
{  
    printf("input temp in celcius pls");  
}  
return 0;  
}
```

Problems Output Debug Console Terminal Ports

```
enter temperature35  
Hot  
PS C:\Users\Huzaiifa\mycode(PF_lab_all\POST_LAB_05> █
```

TASK # 02 — Mobile Data Package (*if-else*)

A mobile service provider checks a customer's remaining balance before activating a package. Write a C program that takes the balance as an integer and uses if-else to display "Low Balance" if the balance is below 500, "Sufficient Balance" if it is between 500 and 2000, and "Premium Balance" if it is above 2000.

```
#include <stdio.h>  
  
int main()  
{  
    int balance;  
    printf("enter balance");
```

```
scanf("%d",&balance);  
if (balance<500)  
{  
    printf("low balance");  
}  
else if(balance>=500 && balance<=2000)  
{  
    printf("sufficient balance");  
}  
else if(balance>2000)  
{  
    printf("premium balance");  
}  
else{    printf("enter correct balance integer");  
}  
}
```

Problems Output Debug Console **Terminal** Ports

```
enter balance700  
sufficient balance  
PS C:\Users\Huzaiifa\mycode(PF_lab_all\POST_LAB_05> █
```

TASK # 03 — Hospital Appointment (*nested if-else*)

A hospital appointment system asks the patient to enter three integer values: appointment, doctorAvailable, and registrationCompleted, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the patient has an appointment, then check whether the doctor is available, and finally check whether the registration is completed before displaying whether the patient can meet the doctor.

```
#include <stdio.h>  
  
int main()  
{
```

```
int appointment,doctorAvailable,registrationCompleted;

printf("\nenter (1) yes (0) no if have an appointment");
scanf("%d",&appointment);

printf("\nenter (1) yes (0) no if doctor is available");
scanf("%d",&doctorAvailable);

printf("\nenter (1) yes (0) no if registration complted");
scanf("%d",&registrationCompleted);

if(appointment==1)
{
    if(doctorAvailable==1)
    {
        if(registrationCompleted==1)
        {
            printf("\npatient can meet the doctor!");
        }
        else
        {
            printf("\npatient canot meet doctor!");
        }
    }
    else
    {
        printf("\ndoctor not avaialable!");
    }
}
else
{
    printf("\npatient doesnt have appointment!");
}

return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
enter (1) yes (0) no if have an appointment1
enter (1) yes (0) no if doctor is available1
enter (1) yes (0) no if registration complted1

patient can meet the doctor!
PS C:\Users\Huzaiifa\mycode(PF_lab_all\POST_LAB_05> █
```

TASK # 04 — Online Food Order (*nested if-else*)

An online food delivery system asks the customer to enter three integer values: restaurantOpen, itemAvailable, and balanceSufficient, where 1 means yes and 0 means no. Write a C program using nested if-else statements to first check whether the restaurant is open, then whether the selected item is available, and finally whether the customer has sufficient balance before displaying the appropriate order status.

```
#include <stdio.h>

int main()
{
    int restauranOpen,itemAvailable,balanceSufficient;
    printf("\nenter (1) yes (0) no if restaurant is open");
    scanf("%d",&restauranOpen);

    printf("\nenter (1) yes (0) no if item is avaialble");
    scanf("%d",&itemAvailable);

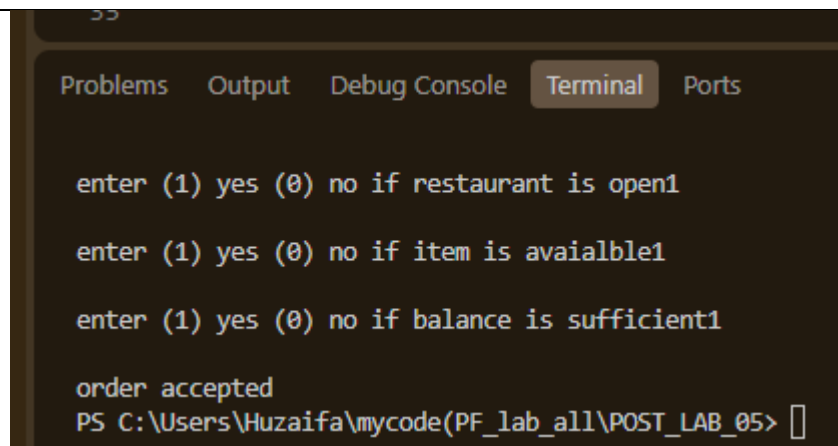
    printf("\nenter (1) yes (0) no if balance is sufficient");
    scanf("%d",&balanceSufficient);

    if(restauranOpen==1)
    {
        if(itemAvailable==1)
        {
            if(balanceSufficient==1)
            {
                printf("\norder accepted");
            }
        }
        else
    }
```

```

    {
        printf("\norder rejected!");
    }
}
else
{
    printf("\norder rejected!");
}
}
else
{
    printf("\norder rejected!");
}
return 0;
}

```



```

Problems  Output  Debug Console  Terminal  Ports

enter (1) yes (0) no if restaurant is open1

enter (1) yes (0) no if item is available1

enter (1) yes (0) no if balance is sufficient1

order accepted
PS C:\Users\Huzaiifa\mycode(PF_lab_all\POST_LAB_05>

```

TASK # 05 — ATM and Transaction Selection (Nested switch):

An ATM provides four operations: 1 for Balance Inquiry, 2 for Cash Withdrawal, 3 for Cash Deposit, and 4 for PIN Change. After the user selects an operation, the ATM displays a second menu related to that operation; for example, after selecting Cash Withdrawal, the user can choose 1 for Savings Account or 2 for Current Account. Write a C program using a **nested switch statement** to take both choices as integer inputs and display the selected operation and account type

```

#include <stdio.h>

int main()

```

```
{  
  
    int operation1,operation2;  
  
    printf("\nenter desired opration\n(1)Balance Enquiry\n(2)Cash Withdrawal\n(3)Cash Deposit");  
  
    scanf("%d",&operation1);  
  
    switch(operation1)  
    {  
  
        case 1:  
  
            printf("\nenter account type (1)Current (2)Savings");  
  
            scanf("%d",&operation2);  
  
            switch(operation2)  
            {  
  
                case 1:  
  
                    printf("current account balance check");  
  
                    break;  
  
                case 2:  
  
                    printf("savings account balance check");  
  
                    break;  
  
            }  
  
            break;  
  
        case 2:  
  
            printf("\nenter account type for withdrawal (1)Current (2)Savings");  
  
            scanf("%d",&operation2);  
  
            switch(operation2)  
            {  
  
                case 1:  
  
                    printf("current account withdrawal");  
  
                    break;  
  
                case 2:  
  
                    printf("savings account withdrawal");  
  
                    break;  
  
            }  
  
        case 3:  
  
            printf("\nenter account type fro deposit (1)Current (2)Savings");
```

```
scanf("%d",&operation2);

switch(operation2)
{
    case 1:
        printf("current account deposit");
        break;
    case 2:
        printf("savings account deposit");
        break;
}
}
```

Problems Output Debug Console Terminal Ports

enter desired opration

(1)Balance Enquiry

(2)Cash Withdrawal

(3)Cash Deposit3

enter account type fro deposit (1)Current (2)Savings2

savings account deposit

PS C:\Users\Huzaifa\mycode(PF_lab_all\POST_LAB_05> █

Repo link: https://github.com/m-huzaifahafeez/M.Huzaifa-Hafeez_26K-3117_PF_LAB-all-